- **First occasion: 10th of September, 2026, 9:00,**
  @F32L2 Physics Student Lab, FIII building 2nd floor

Nanotechnology and Quantum Applications special laboratory 1 – v2.1

Code: BMETEFTBsPSPL1-00
Language: English
Location: BME, EK MFA, Semilab
Responsible: Szabolcs Csonka, András Halbritter
Lab leaders / supervisors:
I András Halbritter (BME) / Zsigmond Pollner (polugyint@gmail.com),
II Sándor Bordács (BME) / Máté Podráczky (cs7b.podraczkim@gmail.com).
III Levente Tapasztó (EK MFA) / Richárd Csanádi (csanadi.richard@ek.hun-ren.hu), Dóra Varga (v.dora04@gmail.com).
IV Dávid Krisztián (Semilab) / Ferenc Korsós (ferenc.korsos@semilab.hu), Gergely Havasi (gergely.havasi@semilab.hu)

Requirements

Attendance requirements: It is compulsory to execute
Option a) 4 measurements or
Option b) 3 measurements plus presentation of own research topic.
Measurements are 2x4hours long. In case of **certified absence**, a maximum of 1 replacement measurement (altogether 2x4 hours) are possible at a time agreed by the tutors.
Option c) like in Option b), except that presence at all 4 measurements, but lab report is required only from 3 measurements.

Working method: the measurements are executed in pairs. Please indicate your preferred pairs for the measurements in this Excel file by 13th September. Option b) is only possible if both members of the pair choose it and a written certificate is required from the research supervisor that the student is actively carrying out research work.

Schedule and place of the measurements (the numbers in the table denote the number of a measurement pair, see the pair codes here):

Time-period of the courses:
Team 1: Thursday 8:30-12:00
Team 2: Thursday 12:15-15:45

A lab responsible will be present continuously during these times. Breaks are not scheduled, but you may take a break and leave the lab as needed during measurements.

Measurements exercises (see in lab descriptions):

- I Neuromorphic circuits - "**Neuro**" (place: F32L2 Physics Student Lab, FIII building 2nd floor)
- II Malaria diagnostics - "**MagnetoOptics**" (place: F32L2 Physics Student Lab, FIII building 2nd floor)
- III Scanning Probe Technique - "**STM**" (place: KFKI Building 26., Ground floor 25., Konkoly-Thege Miklós street 29, Budapest, 1121)
- IV Semiconductor characterization - "**Semi**" (place: Semilab Building Dexagon, Prielle Kornélia street 2, Budapest, 1117)

Team 1 and 2

Week	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Date	9/17	9/24	10/1	10/8	10/15	10/22	11/05	11/12	11/19	11/26	12/3	12/10
Neuro	1,2,3	1,2,3	7,8,9	7,8,9	4,5,6	4,5,6	10,11, 12	10,11, 12				
MagnetoOptics	4,5,6	4,5,6	10,11, 12	10,11, 12	1,2,3	1,2,3	7,8,9	7,8,9				
Semi	7,8	7,8	1,2	1,2	9,12	9,12	3,6	3,6	4,5	4,5	10,11	10,11
STM	9,12	9,12	4,5	4,5	10,11	10,11	1,2	1,2	7,8	7,8	3,6	3,6
Presentation									10,11, 1	3,6,2	4,5,9	7,8,12

Grading:

Measurements

- Entrance test (30 points) at the start of the first session.
- Lab notes (30 points total), submitted at the end of each session.
- Lab report (40 points), due within 1 week of the second session.

Presentation

- Option a) 1 oral presentation at the end of the semester on one of the measurements (100points) in 20 minutes with 10minutes discussion). Evaluation form is used (Scores for quality of a) structuring the content, b) introduction, c) presentation of the methodology, d) presentation of own results, e) adequate conclusion, f) visual quality, clarity of speech, confidence, formal aspects, ...).
- Option b) and c) Like in Option a) plus:
  - One more oral presentation on the own research work (50points)
  - In case the student has participated at TDK conference, no need for oral presentation (50point is automatic) .

Total Points:

- Option a): 500 points
- Option b, c): 450 points

Grade Scale:

- <40%: Fail (1)
- 40%-55%: Pass (2)
- 55%-70%: Satisfactory (3)

- 70%-85%: Good (4)
- 85%-100%: Excellent (5)

Note: The failure criterion applies to the overall score, not to individual components (entrance tests, lab notes, or reports). All measurements must be completed, but a low score in one area will only lower the aggregate grade without requiring a replacement measurement.

Submission Rules:

- **Entrance Tests:** Must be completed in the class in hardcopy. Late arrivals will have less or no time, leading to a potential loss of points.
- **Lab Notes:** Submit online in Moodle within 1 hour of the end of each session. Late submission results 10-point-deduction per hour.
- **Lab Reports:** Must be submitted within one week (7x24 hours) from the start of the second lab session. A one-time, one-week extension is allowed without penalty. Any further delay results in a 10-point-deduction per week, until the score for the lab report reaches zero.

Lab leaders should correct the lab reports within one week from the submission deadline, with a single possibility for an extra week delay. Please complain, if this is not satisfied.

Student Pairs:

It will be available on Moodle.

Oral presentations:

In case you chose Option B or C:
- you have to prepare 2nd talk about your own research - or prove that you presented it at the TDK conference.
- in case you had no TDK presentation, please bring a written certificate from your supervisor that you are working with him/her.

Places

Semroom - Department of Physics, F building I, 1st floor, Room 10

Program of oral presentations

It will be available on Moodle.

References to prepare your talk:

Neuro

[1] F. Aguirre, et al. Hardware implementation of memristor-based artificial neural networks. Nature Communications, 15, 1974 (2024).

[2] H. Liu, et al. Artificial Neuronal Devices Based on Emerging Materials: Neuronal Dynamics and Applications. Advanced Materials 35, 2205047 (2023).

[3] Y. Huang, et al. Memristor-based hardware accelerators for artificial intelligence. Nature Reviews Electrical Engineering, 1,

286 (2024).

[4] W. Yi, et al. Biological plausibility and stochasticity in scalable VO2 active memristor neurons. Nature Communications, 9, 4661 (2018).

[5] O. Maher, et al. A CMOS-compatible oscillation-based VO2 Ising machine solver. Nature Communications, 15, 3334 (2024).

Magneto-optics (Beside Ref. 1, please choose at least one more)

[1] https://en.wikipedia.org/wiki/Jones_calculus

[1*] more details: R. M. A. Azzam and N. M. Bashara: Ellipsometry and Polarized Light (North-Holland: New York, Amsterdam and Oxford, 1977.) Chapter 1.1-1.6 & 2.1-2.3

[2] Butykai, Á. Orbán, V. Kocsis, D. Szaller, S. Bordács, E. Tátrai-Szekeres, L. F. Kiss, A.

Bóta, B. G. Vértessy, T. Zelles and I. Kézsmárki, Malaria pigment crystals as magnetic

micro-rotors: key for high-sensitivity diagnosis, Scientific Reports 3, 1431 (2013).

[3] Kikuchi, H., Yokota, M., Hisakado, Y. et al. Polymer-stabilized liquid crystal blue phases. Nature Mater 1, 64–68 (2002).

Outlook to magnetic textures and their imaging using polarized light [4] and electron microscopy [5]

[4] A. Hubert and R. Schäfer: Magnetic Domains. The Analysis of Magnetic Microstructures. Springer Verlag, Berlin (1998) Chapter1, 2.1 & 2.3.

[5] Chapter 2.4 from Ref.2 + Yu, X., Onose, Y., Kanazawa, N. et al. Real-space observation of a two-dimensional skyrmion crystal. Nature 465, 901–904 (2010).

Semi

[1] D. H. Macdonald, et al. Iron detection in crystalline silicon by carrier lifetime measurements for arbitrary injection and doping. J. Appl. Phys. 95, 1021–1028 (2004)

[2] T. Pavelka, et al. Carrier Lifetime Measurements in Silicon for Photovoltaic Applications. 2007 ECS Trans. 11 331, DOI 10.1149/1.2778675

[3] D. Krisztián, et al. Improved accuracy of eddy-current sensor based carrier lifetime measurement using laser excitation. EPJ Photovoltaics 13, 3 (2022), https://doi.org/10.1051/epjpv/2021014

[4] D. Krisztián, et al. Carrier Lifetime Measurements of Monocrystalline Silicon Ingots Using Upgraded Laser E-PCD Technique. EU PVSEC 2021, 10.4229/EUPVSEC20212021-2DO.10.6

STM

[1] Binnig, G & Rohrer, H. Scanning Tunneling Microscopy – From birth to adolescence, Nobel lecture, 1986. December 8.

[2] Binnig, Gerd & Rohrer, H. In Touch with Atoms. RevModPhys 71 S324 (1999).

[3] Andrei, E.Y., MacDonald, A.H. Graphene bilayers with a twist. Nat. Mater. 19, 1265–1275 (2020).

[4] Wiesendanger, R. Scanning Probe Microscopy and Spectroscopy. Methods and Applications. Cambridge University Press (1994).

[5] Meyer, E., Hug, H. J. & Bennewitz, R. Scanning Probe Microscopy. Springer-Verlag, Berlin Heidelberg (2004)

+[1] S. Hembacher, F.J. Giessibl, J. Mannhart, & C.F. Quate, Revealing the hidden atom in graphite by low-temperature atomic force microscopy, Proc. Natl. Acad. Sci. U.S.A. 100 (22) 12539-12542, (2003).

Guildlines to prepare your talk

1) About the structure of the presentation:

On the **title** slide please include your name, the date, and your affiliation / place of research.

Start either by describing the **motivation** to the research, or a contents slide. Show a list of **contents** at some point in the presentation, either before or after the motivation, but definitely before you get deeper into the topic or your results.

The motivation can be as short as a couple of minutes, but don't be too short. It can be as long as 4-6 minutes (depends on the topic and your time limit), but stick to a few things only. It is counter-productive to present lots of examples and lose focus. If you mention something, make sure you know the background at least superficially (look in the references if necessary).

Discuss your experimental **methods** and the theoretical **background** at least in passing. If it is complicated but the details are not important, you can skip parts of it and focus on what is relevant later. If they are important, spend at least a few minutes on the basics and also define the physical quantities, so that the audience has some concept of it all.

Structure the talk in your head, and present it accordingly. For example, pause between major parts for a few seconds and then say out loud what comes next. You could even put in an extra slide before each main part of the talk that shows only the title of this next part, in line with the Contents.

At the end, make a **summary** or conclusions slide to go over your main results again. You can even show earlier pictures again. After the conclusions you might mention again some of the practical applications, but do not discuss them in detail. Focus on your achievements, stay positive, and maybe give an outlook and talk about how you plan to continue your research.

Finally, **acknowledge** all important contributors, from fellow researchers who took part to other research groups who provided materials, or institutions where you performed part of your research, or funding projects, grants and fellowships.

2) General advice:

Practice the talk at least a couple of times out loud, and measure the time. Cut off parts as necessary to keep it short enough.

Use **figures** and illustrations, try to avoid slides with text only, especially in the motivation / introduction. If possible, do not describe something and make the audience rely on their imagination, if you can instead illustrate that something by a

schematic or a photo. Don't leave half a slide empty (unless it is a title slide); add an illustration or enlarge an existing one (within reason), or merge two slides into one if it is reasonable. If you have data in text or in tables that can also be shown in a figure, make the figure. If a schematic seems old/ugly, you can find a newer one or make one yourself.

Note on the slides the sources and references that you get your material from, especially for images and figures. Add slide numbers.

Please don't use screenshots of computer programs (windows and other user interfaces), at least crop to show only the graph.

Define quantities on the slide: actually write down their names.

Don't write too much text if possible, like long sentences. No need to write down every numerical result. No need to present every piece of research. Focus on a main story and do not get too sidetracked.

You may use notes but don't rely on them too much, otherwise you may get stuck easily. Make the slides so that they are helpful for you, then remember to look up at your slide. And we can turn this around: describe your figures during your talk, if it's up there, you should explain it or just point to it at least once. Use the pointer/mouse.

In evaluating the data, always try to consider what may be the error of the quantity, at least from how much variation there is between repeated measurements. Then consider how many decimal digits in the quantity are useful. For example, if the quantity you read out is 2.0578 and the error (or scatter) is 0.1579, the useful digits in both are roughly two, and then just write 2.06 +- 0.16. Don't estimate relative differences between quantities that are themselves relative: like temperatures in Celsius (it is not an absolute temperature scale, but measured relative to 273 K).

You can mention challenges or failures, but focus on how you overcame them. You can mention if there was something that did not work or that you could not finish, but do not dwell on it and do not collect them in one place. Focus on the results that you have achieved, and try to **stay positive**.